# EXPERIMENT 1(A): $TE_{10}$ MODE SIMULATION.

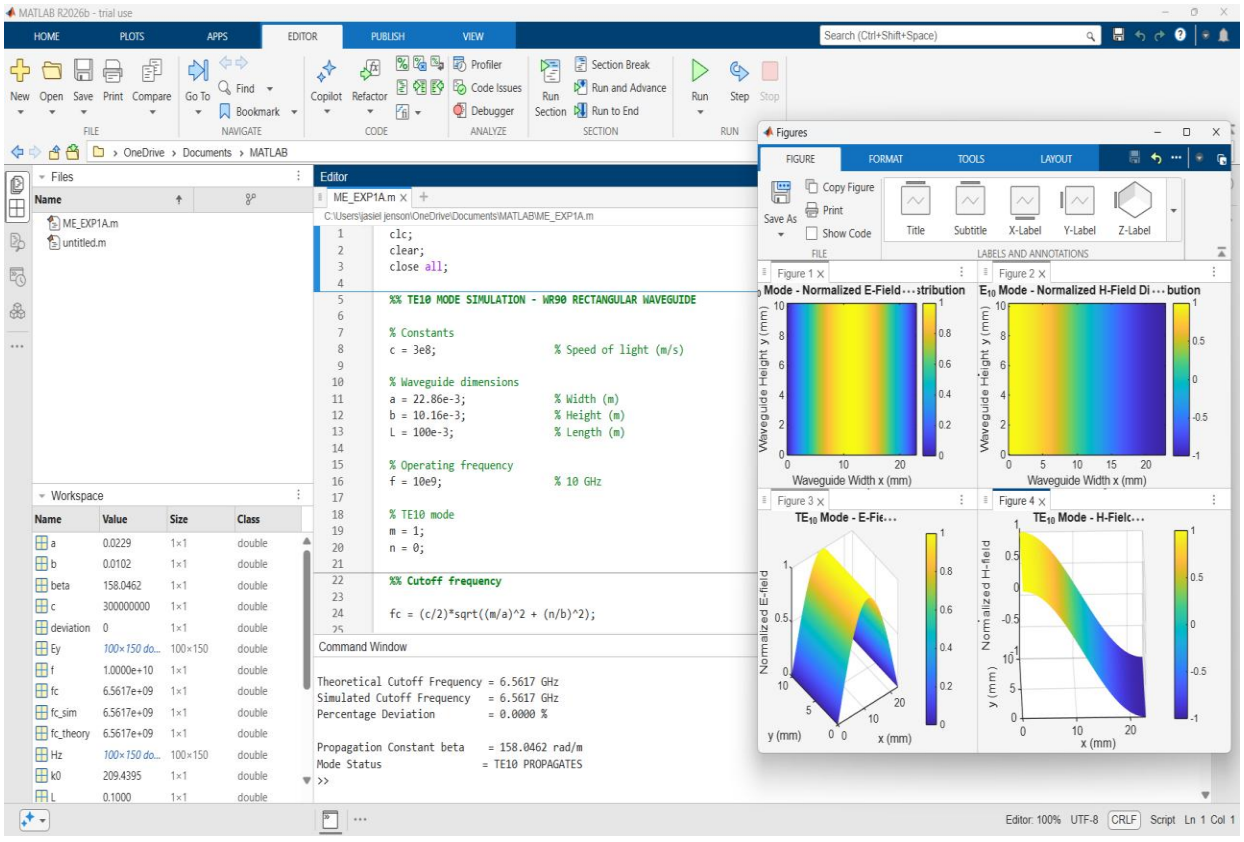
## Aim:

To simulate the $TE_{10}$ mode in a WR-90 rectangular waveguide and analyze its electric field, magnetic field, cutoff frequency, and propagation constant at 10 GHz.

## Procedure:

1. Define the waveguide dimensions: \(a=22.86\) mm, \(b=10.16\) mm, and \(L=100\) mm.
2. Set the operating frequency to 10 GHz and select $TE_{10}$ mode.
3. Calculate the cutoff frequency and propagation constant using theoretical formulas.
4. Plot the electric- and magnetic-field distributions using MATLAB.
5. Compare theoretical values with simulated values, if available.

## Output:



## Result:

The $TE_{10}$ mode was analyzed, and its field distributions were plotted. Since the operating frequency is above the cutoff frequency of approximately 6.56 GHz, the mode propagates through the waveguide.